

ActInf Livestream #013.2: "Cybernetic Big Five Theory with the Free Energy

Livestream | 2:09:26

all right hello everyone

welcome to the active inference live

stream it is active inference live

stream 13.2

on january 12th 2021

today is going to be a really awesome

discussion so thanks everyone who has

come out

to participate welcome to the active

inference lab

we are an experiment in online team

communication learning

and practice related to active inference

you can find us at our website

twitter gmail youtube or our keybase

public team and public username and this

is a recorded in an archived live stream

so please provide us with feedback so

that we can improve on our work

all backgrounds and perspectives are

welcome here and as far as video

etiquette for live stream goes

mute if there's noise in the background

raise your hand so we can hear from

everybody who wants to speak

and we'll use respectful speech behavior

all the active streams in 2021 at least

the regularly scheduled ones are on

tuesdays from 7 to 9 pacific time so

check out this

rb.gui link to see the schedule

today we are in 13.2 on january 12th

with adam saffron and colin deyoung

talking about

this cool paper about cybernetics and

personality and next week and beyond
we're going to be heading into
different sets of papers so check out
this spreadsheet and see if you're
interested in participating
um douglas i'm retired from
information technology all right i just
had to mute
douglas so yeah please don't play a
stream while you're on a live stream
this is a
cognitio 2021 and this is a conference
in 2021 that is being organized by
several of our colleagues
uh by mao and bayern s who has
appeared on the discussion before
welcome blue and so this looks like a
really cool
conference that they are all organizing
so
check out this conference if you're
curious and hopefully we'll be able to
have updates on this conference
later on so in the intro
in warm-up session we introduce
ourselves we can just give a short
introduction or warm-up and then pass it
to somebody who hasn't spoken yet
so hi everyone i'm daniel i'm in
california
and i'll pass it first to blue
hi everyone i am blue knight i'm an
independent research consultant based
out of new mexico
oh and i'll pass to steven
hello i'm stephen i'm based in toronto
i'm doing a practice-based phd
at the southern school institute of

applied psychology

and i will pass it over to sasha

hi i'm sasha i'm based out of davis

california

and i'm a neuroscience graduate student

and i will pass it to colin

hi there i'm colin de young i'm a

professor in the

psychology department at the university

of minnesota

and i will pass it on to adam

hi i'm adam saffron i'm a postdoctoral

researcher

out of bloomington indiana

good to be with you ooh go to alex

yeah hi everyone i'm alex i'm in moscow

russia

and i'm researcher in systems management

school

and i texted to scott

thank you hello everyone my name is

scott david i'm at the university of

washington

applied physics lab where i'm the

director of the information risk

research initiative and i'll pass it to

marco

hi thank you my name is marco arcolin

i'm from holland i graduated in colorado

science and

i'm just very very interested in active

infants i'm not affiliated at the moment

but i'd like to be here um is there

anyone left

okay i had to kick out somebody who's

doing feedback so please have it figured

out

and yes cool all right

maybe not all right hello
uh dave are you there
okay he just got disconnected i am here
hello hello
uh okay well just go just proceed i'll
figure it out
fun great times thanks for being on so
for the warm-up questions people
can just raise their hand if they want
to say something that's
related or unrelated uh the same
questions from last week will put up and
take a little bit of a different go
at them so what is something you are
excited about today
and what is something that you liked or
remembered about the paper
uh anyone can go for it i'd be curious
to either of the authors just
looking back on it it's not so long ago
but we are in a new
year so with the fresh lens of
the numerology of the year and just
maybe these discussions
how are you looking back at what you had
done
and seeing it in a new or a different
way
or any other participants who want to
add something that they
thought of go ahead
oh i i might as well go ahead then i was
just going to say that i don't know that
for me maybe even more so than for adam
this particular chapter was sort of
right at the boundary of the things that
i've been thinking about and
working with adam has really pushed me

into um
you know making contact with this whole
topic of active inference and the kind
of
you know fristonian world um and then
it's also made me
think more about about
modeling um because i've generally been
fairly skeptical
about the value of trying to model whole
agents that are analogous to actual
human beings
uh for learning things about psychology
just because uh
because the brain and people are so
complex you know if you
if you have a model it has to have a
sufficient degree of verisimilitude
in resembling the thing that you're
modeling before you're likely to
actually learn new things just
from studying the model rather than
learning about you know the limitations
of the model in one way or another and
your skillet modeling
but i guess one of the things that
writing this chapter with adam
made me think about was mars different
levels of analysis
and the way that if you get a good
enough uh resemblance
at the level of uh of function or
computation
you might be able to learn things even
if uh
you don't have verisimilitude at the
level of
implementation uh or or algorithm right

we don't necessarily have to say that the brain is doing the exact kind of mathematical computations that that you know fep demands and in the same way that we would do them to model it but as long as we're capturing something that truly resembles the functional computation that a cybernetic system like a human being needs to carry out we might actually be able to learn new things about the way that people work

cool steven and then anyone else who wants to raise their hand but just maybe we could hear from everybody in the warm-up so steven and then adam and then anyone else

i'm really excited by the fact that this is a you know taps into a very widely used tool in psychology so and then links this back to these new potentials of active inference which you know we've talked a lot about maybe in more theoretical philosophy philosophical ways but this really is showing um how it you know the twos can start to move into each other so i'm curious today to see more ways in which you know there's questions about um you know the potential for that going forward you know what what possibilities in the field of even organizational psychology which

uses these traits as well it's this
these these traits are
kind of used beyond kind of
work with um sort of therapeutic
environment so um and you know even with
the myers-briggs and other types of
things how can
a more dynamic or kind of embodied you
know what's the future basically
for applied psychology who
adam then scott
um so this
actually working with colin on this um
it really stretched me
also was that the like limits of what i
was used to thinking about
and uh i think i never
like really seriously engaged with
personality science before this like i
was always
interested but then i think i was also
like skeptical that
they had any uh chance of carving nature
at his joints
and i was just like uh maybe and
but then as we started exploring the
correspondences between like the
cybernetic view of personality
and the other cybernetic perspective of
active inference i started to become
more compelled
and then i'd say like things are always
kind of moving around for me in terms
of like uh
perspectives but um i think recently
some of the things i've been thinking
about is
like how far can we

stretch or apply this these ideas of
personality
over uh what scales and what ranges of
systems uh
can we think of the personalities of
societies the personalities of
uh any system uh and so
that's i think most recently been uh on
my mind
but i'm excited to be here because uh uh
i love these live streams and uh last
week was awesome
cool agreed scott and then sasha than
anyone else
so a couple of things the the the active
inference is totally new to me
and it's really been revelatory as other
people have said
a couple of things i used to work in
identity a lot still do
identity kind of systems federated
identity kind of technical stuff
not on the technical side what i
realized is the attributes are carried
in these
markov blankets right so the you know
the the access to attributes is through
this
the statistical views of the surface of
other systems
and so that's been really useful to me
and then the other part i realized is
i realize that a lot of folks um fristen
and others are coming at this from the
biology perspective of seeing
gee how do these things emerge by
themselves what i realize is as a lawyer
all i've ever done is construct

synthetic markov blankets
because contracts are revealing a
surface of a system in fact
the system doesn't exist until the
contract exists forget
corporations if i want to put a supply
chain in place i write a contract and
then
the organism of the supply chain exists
as a result and so it's funny because in
that idea of
if it's too rarified for viewing human
mind i get it
and it's interesting because in the
de-risking context where a lot of folks
are getting together
they try to make it simpler you know
worry about bankruptcy worry about
notice worry about these specific things
boom boom
that's all they care about so it's kind
of like a synthetic
organism constructed by the contract the
markov blanket defines the organism it's
like the opposite of the
blind men and the elephant where they're
trying to find the elephant here there
is no elephant until there are other
parameters
so the the attributes define the system
and it's a very bounded system very few
attributes to it but it simplifies it
and that's what a lot of regulatory
market issues are about so that's been
huge for me
really realizing that we can
synthetically construct these systems
and base them on shared story shared

narrative

shared meaning contracts as a formal way
of doing that but folkways is an
informal way of doing that and i think
that's absolutely a very rich
avenue for for my work and i'm
looking forward to working with others
on it thanks cool

thank you sasha and then anyone else on
the warm-up

yeah sasha then adam okay um

yeah uh i'm just really kind of

following up on what adam

mentioned um thinking about other
applications for

um the big five theory and what other
groups and organisms they might apply to
and i think uh

very relevantly for this group and
of the kind of projects that we're
working on is um

personalities of a team um and uh
thinking about different behaviors i'm
i'm sure this isn't a novel concept um
but i'd love to hear more about it from
the authors

of other possible applications

um of these metrics for uh teams
of varying sizes um and the other
thing i've been thinking about since
last week is this

um if this paper aims to integrate
these two concepts of active inference
and

um the big five theory um

are there parts that are incongruent is
there something that doesn't

um that can't work together from both of

these systems
and is there something that active
inference
can glean or kind of take away
from personality research in a way that
you know hasn't really been tried before
cool questions thanks adam then blue
uh somewhat flipping uh what i said
before related to
uh what sasha and scott just said um so
like in
also not just applying personality to
other systems where you might not
normally think of them in terms of
personality
um i'm also interested in like reversing
that flow
so it's like we how might we think of
the personality of teams but
the things we know about teams how might
that influence how we think about
persons and their personalities or uh
contracts like what are analogies or
models that could be derived
by thinking of what you described or
this like inactive
contract like creating the mark of like
the contract
in the in the doing
um to what extent can we
have this as part of um person formation
and personal evolution personality
evolution um i don't know if that makes
sense but
that's another place i've been thinking
is like this bi-directional flow of like
taking things from persons putting them
in terms of things we might not normally

think of as persons
taking things we might not normally
think of as persons and then
putting it back to persons and their
personalities yeah it's kind of like
finding the human in the non-human
atom really interesting point so
personifying and
enlivening things that are perceived as
inanimate and then finding the mechanism
inside of living systems² so blue then
colin so going back
to what adam was saying not just now but
earlier um and
also just something that i liked about
the paper so for me psychology has kind
of always been like
this soft science like you can't really
do science on psychology but really like
what what that means is that it's a
wicked problem right
it just falls into the complexity realm
like people are
super super intricate complicated it's a
convoluted like thing
so this paper for me really um was
awesome in the way that it kind of
concretized
or hardened up for for me like a
quantitative way to
um really look at personality
interesting colin than anyone else and
i'll put up the third question as well
well first i'm really glad to hear that
this might have convinced somebody that
psychology wasn't just
soft science uh it's it's definitely
hard

science in the sense that it's difficult
um
and that's largely about you know issues
well it's two things i think it's issues
with measurement
um the the kinds of measurements that we
want to do and can do
you know they don't necessarily line up
very well in terms of precision and so
you know everyone has physics envy
because you've got precision down to
10 decimal places on measurements but
you know we're lucky if we have
precision down to uh down to one decimal
place or if it's even clear that our
measurements are
you know anything other than essentially
heuristic you know if you have a
if you have a scale and you're measuring
a personality trait
and essentially what you've got is just
a measure of people's
uh you know degree of a particular kind
of behavior or experience
relative to other people so the
measurement is relative and it's not
clear that the
you know one point on the scale actually
corresponds to
uh some true interval
rather than just a kind of heuristic
approximation so yeah there are a lot of
really difficult challenges in
psychology
but nonetheless i mean in order to do
science you just have to have some kind
of standardized measurement
and then be able to uh to replicate

right to see if other people can get it
i used to think that you had to have uh
you know hypothesis testing too but then
i realized you can really just do
science
if you do it exploratory work and then
replicate it
because once you've found something that
then de facto becomes the hypothesis for
the replication attempt
so all you're really looking for are
reliable patterns
in reality fundamentally and i think we
can do that to some extent
in psychology i wanted to say something
also about uh
sasha's question about the
you know whether there are
incompatibilities between
um fep and active inference and
um general approaches to
personality and my theoretical approach
specifically
um and i think one of the things that's
really nice about active inference is
that
it's uh it's it's very low level in some
sense it's this formalism
that is extremely widely applicable um
and the way that i see it is that it's
essentially a formalism
that captures the essence of all
cybernetic systems
um and so you know i haven't found
anything
that makes me say oh this isn't gonna
work you know it's like
there's a lot of detail about what

people are
that isn't captured in active inference
but only because it's so
low level but like there's nothing about
active inference that isn't crucial to
what people are right because
you know that the process of uh
you know pursuing goals via feedback
that is just necessarily the core of
what an agent is
and so i guess my concern isn't that
they're incompatibilities
it is how to draw out the ai framework
in such a way that it makes contact with
a lot of the
specifics that we know about humans
function rather than the generalities
about how all cybernetic systems
function
really interesting answer we'll go to
dave
and then markup hi my professor
of cybernetics gordon pasque in the
1950s and 60s
developed some very fine-grained
measures
of learning strategies
which is how individuals approach a
specific concrete task as they're
learning it
and learning styles which is a more
systematic
way that folks uh develop
a fixed repertoire of approaches to a
task that
the extremes of the on that scale are
serial learning and holistic
learning i believe the

the areas of his data where he uses
uh entropy as an important measure for
certainty and for tolerance for
ambiguity
should tie in with um
the um active what
the not not the active insurance the
other the other major area
that we're talking about uh free energy
and i suspect that the
learning styles should correlate with
personality and that is an area that i
believe he
uh past did work on so that's something
i need to dig into the literature on and
compare it cool marco
hi thanks um also thank you scott that
was very well put
um so
in general i just really like a lot of
things that have been said um
about active influence as a as a
framework i
i also echo the notion that the beauty
is that it's so
uh generally applicable that you can
kind of elaborate it and
that becomes a kind of game of if you
take a bit further with a few
assumptions
uh when does it touch or overlap or
align with other existing years or
empirical data
but what has been mentioned and
something i find personally an
intriguing possibility
is um not necessarily active inference
but if something like active influence

would be true
and it would indeed be the most
approximate description
of our neural dynamics then
naturally follow that it also can become
quite intuitive so just like colin said
that you can elaborate
until it kind of touches something else
in terms of theory you could also maybe
consider the idea that you can elaborate
your understanding of active influence
relative to personality or other
psychological notions
until start to touch your kind of social
psychological notions
of people of yourself of others society
and i think that's what i think someone
in the beginning said i think i think
was stephen about the future of applied
psychology and i think this is really
really crucial
not just from a scientific perspective
but also a social and
humanist perspective because
i mean i think most people agree that
folk psychology isn't enough and i think
also most people agree that most
scientific psychology
whether it's a neuro or clinical side
isn't very easily implementable
uh in terms of actual understanding
yourself by first person
so i think that's definitely the most
exciting aspect i think this
particular work is a very nice milestone
with regards to that
i mean i don't think anyone really
understands itself that much better with

uh some ratings on the big five but
knowing the mechanisms i think is a very
very beautiful lens to understand myself
so
cheers thank you for sharing steven and
then anyone else
yeah so following on from what marco was
saying though is
like the the fact that this paper also
takes in the affordances
and and active influence like you say
does this dynamics you
you gives this generative potential to
the
big five which i know is kind of
sort of seen to be there but you know
the the challenges that they can end up
being
whether you like it or not prescriptive
and kind of diagnostic
to some extent and having a way to
relate them
opens up that phenomenological sense
because there's a i think what i'm
really interested is
if people structure their metaphorical
models of how they
cognized in the world like how they
whether they relate to time as time
coming towards them or they're going
towards time or whether
they're um trying to um
build a a more firm career
or they're trying to get on in the world
and travel and go
somewhere so those kind of
phenomenological ways of structuring
thought

they they seems to be more ways that you
can then connect them
as opposed to it being like completely
disconnected
or just too big a gap to bridge so this
uh
this mixture of the affordances the
dynamics the blanket and the attention
and the way that phenomenological
um sort of structuring could happen even
at the individual level
starts to become more plausible which is
kind of cool
awesome yeah a lot of cool ideas coming
up
one theme to return to is that
by looking at the patterns and thinking
about the frameworks that span
different systems of different levels of
analysis we can think about
like the traits of organizations
teams individuals organs cells so that's
the multi-scale
kind of complex systems approach because
we set out
to look for patterns across systems and
then that led to all these interesting
questions like would there be a big five
for teams or how are the generative
models of individual personality
related to the generative models or the
generative processes underlying
group personalities and so pretty
interesting things that we're coming to
any other points
in this intro before we just return to
some of the later pages
colin go ahead and then anyone else i'll

just make one point in relation
to this issue of uh you know team
dynamics and personalities at other
scales
um that's something that
there are different ways to approach it
right so people have certainly
psychologists have certainly studied
uh what happens to teams that have
different
characteristics of the individual
members
uh and so you could think of how do the
personalities of individual members of a
team
influence the success of the team
um so for example people having people
high in openness on the team
tends to uh make it more effective in
various ways and it seems to be because
of
that's especially true for leaderless
teams right so if you have a it's not
necessarily true if there's a clearly
established kind of leadership structure
in the team but if you have leaderless
teams then having people who are more
sort of cognitively
flexible enables them to kind of
organize themselves better so
but i think that what we're thinking
about here is something a little bit
different like is there a personality of
a team as a whole that might be more
than the sum of its parts
um and you know they're
you're probably they're probably going
to be well you you need to do a sort of

parallel kind of research in
terms of trying to come up with a
comprehensive
just set of descriptors of the
properties of teams
and then seeing if they varied
systematically in the same way that
people did with individuals and
leading to the big five but i guess one
of my intuitions
is that the highest level of the
personality hierarchy
stability and plasticity which are also
the most replicable across different
languages when people have done the kind
of big five research
so there's this very clear way in which
uh what i think
of as the two fundamental properties of
all cybernetic systems you know
at least ones that have to be able to
function in an unpredictable and complex
environment
that is probably applicable at virtually
any scale even for something that might
not be well
it's i think it's an interesting
question whether something like a team
or a society
is actually a cybernetic system itself
or whether it is an emergent dynamic out
of a set
of cybernetic systems interacting with
each other
but regardless i think you can certainly
describe the stability and plasticity of
teams or even societies right you can
describe

them in terms of how much they focus on
and value

the lack of disruption of their
functioning versus how much they value
expansion and exploration and dynamic
growth

cool that reminds me so much of the ant
research and

do colonies have phenotypes or is it
just the phenotypes of nest mates
to selection shape colony traits or is
it just shaping the nest mates and
they're duking it out

so all these multi-level systems we kind
of can see these similarities
so adam and then scott and then anyone
else

so in relation what collin just said
um correct me if i'm wrong but i believe
in terms of like cross-cultural
like the robustness um like
is it the stability and plasticity like
this like that level of granularity
actually is what is

the most broadly applicable in terms of
like showing the least

variation like if you look at it in
those two yes so in terms of the lexical
research like starting from a specific
uh language and extracting descriptors
and then doing the same kind of
you know ratings and factor analytic
work the only truly universal
solution that's been found is the
two-factor solution and those factors
very closely resemble

stability and plasticity okay so yeah so
that would support the idea like if

we're trying to do like a multi-scale
personality story or a
quasi-anthropomorphizing of
of uh of quasi agents or maybe we should
just call them agents
depending on our view that yes we will
be most likely to find stability in
plasticity
but um you know because like you
basically grounded the big five
as part of cybernetic functioning and if
any uh cis like from an active inference
perspective if any like
sustaining uh system is doing some kind
of
either explicit or implicit goal pursuit
we actually
might find like even the big five itself
like being surprisingly applicable
although like as you're saying like it's
an empirical question
and so like so i guess the idea is we
actually do the research program of this
sort of
derivation of factors and and with and
then we should expect to find things
that
might be uh shockingly like the big five
and places where you might not expect
them
nice i didn't raise my hand but i just
wanted to come in and say like i think
the best analogy for that that's going
on in science right now is
uh over the last let's say 20 years
people have really started to take
seriously the idea of personality and
other species

now i mean whether personality is the right word there or we want some other one it doesn't really matter right you have individual differences in psychological functioning of other species and so people have been doing this attempting to come up with you know extensive sets of descriptors of other species and uh you know ratings of them or coding based on you know video or observation or whatever and then looking to see whether there are these similar patterns and most of the big five are identifiable uh or you know fairly close analogs in other species and um the one that is the least commonly found is actually maybe sort of uh you know proof of concept in a way because conscientiousness is the dimension that least appears in other species something like conscientiousness and and in fact the only two species where something close to that has appeared as its own separate dimension is in bonobos and chimpanzees right and so you think like what really makes humans special well it's our ability to plan for the distant future and to prioritize goals that aren't uh me immediate or near immediate and to really have these elaborate goal hierarchies

and so it might not be surprising that
it's only our nearest evolutionary
neighbors
that have these kind identifiable
patterns that are
uh that are related to those abilities
and that are
separate from things just like you know
how emotionally reactive you are
very interesting scott and then i think
adam and then marco
so that this is great this um one of the
things i thought was interesting is like
you know
what gets measured gets done and what
gets measured gets seen
and so it's kind of interesting because
we're coming it's like a bio theory of
everything
kind of idea here and we're like wow
this is really explanatory
my recollection is that quantum
mechanics was just a calculation
convention
originally but it kind of works and we
kind of use it and even though we don't
understand all of it
it's kind of helpful that way so it's
not so much is it true but is it useful
and it's this is broadly useful and so
it's interesting in living systems or
cybernetic systems
we start to get at these questions we
ask and when we ask questions from the
i don't know i'd like someone to tell me
what the big five is after i make this
comment but
the um the if they're the big five of

consciousness then
the um i always think of them as the
accounting firms but that's just because
i was a lawyer for so many years
but um but the uh what are
we it's like we're using the turing test
in a way it's too much hubris about
our form of intelligence right and so if
we point our
measurement instrument at a certain
thing that's we're going to see if we
point our measurement instrument at
human
cognition human consciousness our sense
of human consciousness
that's what we're going to see and so
this is a great discussion there's no
it's not a to me it's not a
truth-finding thing but a
a usefulness of the paradigm kind of
issue and if someone could tell me what
the big five are just list them that
would be good because i'm not familiar
with the term
as in parlance sure it's the big five
dimensions that are empirically observed
when you ask people personality trait
surveys
and their openness conscientiousness
extroversion
agreeableness neuroticism ocean and then
um
one metaphor on the big five and uh it's
kind of like
the big five continents that's what we
have in the world
as we have it and so that is specifics
that's related to history and that

changes

but the the principle is water and land
and so maybe in a different system a
different sphere with a different type
of water level

you still would have water and land but
the actual number
of big dimensions might be empirically
seen to be different

so that's kind of an interesting thing

um i think adam

then marco then colin

uh i don't know if this would be useful
but perhaps in a bit actually might be
good to

have colin talk about like the specific
mappings of the big five

onto uh cybernetic function that he
describes in his original paper

uh that could be useful for people but
um

it's

interesting about um so conscientiously
being less likely to

generalize potentially across species or
systems

because uh it seems like that makes some
sense to me and that

um in terms of preserving uh goal
hierarchy integrity

as you're confronted with all these
different like circumstances but you
want to maintain this overall ex like
broadly applicable gold hierarchy that's
benefiting you overall

um in order for that to be the case you
have to have

a system capable of generative modeling

with or modeling with a certain degree of temporal depth and the policies it's pursuing and counterfactual richness such that it could actually get itself into trouble and get confused and go down like wrong paths and so you wouldn't expect this for all systems you would expect this you need a certain degree of sophistication even maybe sophisticated effective inference in order for you to really actually need something like conscientiousness so that one actually might not generalize as much um the and related to what scott just said um there seems to be this yeah interesting combination it's like um realist versus instrumentalists or it's like you can think of like this personality is almost like uh like the big five is like an effective field theory over persons um where it's um a good approximate description um a a useful course screening but now the question i would have though is but is it like just completely so there's a sense in which it's modeler relative it's based on our interests but there's like another sentiment which it's like well um like what worked in evolution and like it's like you don't have to take the perspective of evolution but it's a we can all kind of converge

on it's like quasi objective like
it's one of many perspectives but it's a
pretty darn good perspective
or an important one to keep in mind and
so in that sense the big five actually
might be
carving nature at its joints in a really
powerful way and so i guess one
more thing along those lines is um uh
can be good for probably column to talk
about this since you discovered them
uh which is the aspects um and so one
thing i'm wondering is
uh so it seems like in a lot of
studies when you do things at the level
so
as a correct me if i'm wrong but it's
kind of
um so under so the big five has this
hierarchical structure
over them you'll have these two
metatrades which we've been discussing
of stability as shared variants of
conscientiousness agreeableness and
inverse neuroticism
and plasticity a shared variance of
extroversion and openness
but then underneath the big five um
column
that you'll get if you like force if
you'd like to have the water level uh
go there's a down or up
i'm i'm messing up daniel's like analogy
but you'll get
um two and only two
aspects underneath each of them and
what's interesting is they
oftentimes kind of read as being across

purposes with each other
it's like these are very different not
across they're very different it's very
different to be like
industrious versus orderly like these
are very different ways of being in the
world or if like you're neurotic
it's like very different to being like
um withdrawing or being like more
volatile and like reactive
like um and so i guess one thing i'm
wondering is like
both in doing um active inferential
modeling of
persons and trying to suss out
personality
would it be better to actually in some
ways focus on the aspects
in terms of finding joints there but
also maybe across
other cybernetic systems like could
there be more purpose
uh purchase from the aspects who and i
just flipped to the slide so just
for anyone can just say go to this
figure or go to this other thing so
colin and then marco
sorry you're muted
thank you for the reminder um yeah so
while there's a lot to
uh think about in what's been said in
the last
couple of comments now um
so in terms of this idea of uh
of continence and water and land i do
think that there's a way in which
uh you know stability and plasticity are
like the equivalent of

of water and land so that pretty much uh
any organism
you're going to be able to identify you
know if there's any kind of individual
variability which of course there must
be in order for evolution to proceed
like so probably um you know if you take
amoebas for example they're not going to
have a lot of different personality
dimensions
but nonetheless you might find some that
are a bit more
of the same species that are a bit more
exploratory and others that are a bit
more
uh more cautious let's say so
you know even at this very simple level
you may get this trade-off between
uh do you do things that keep your
functioning
running uh without disruption or do you
do things that lead to
greater uh expanse and uh
discovery of new resources so those
probably do trade off
i would hope that when we're within a
species though that the the big five are
not
um like just the continents in other
words they're not just a historical
accident i think that they actually
reflect
variation in parameters that have
evolved and that are present
in uh in every you know intact
organism of that species
the issue though in terms of carving
nature at its joints is just that the

nature in this case is so complicated
right so
uh if you look at this is a good figure
actually that
uh daniel put up to make this point
which is that
um when we draw these kinds of
hierarchical taxonomies of personality
traits
uh they are necessarily over
simplifications
especially at the levels below the big
five because we know
that personality does not have simple
structure
and what that means from a factor
analytic perspective is that
if you took the taxonomy uh exactly
literally that's depicted there
you would say well for example uh no
trait that is underneath
extraversion that's a subcomponent of
extroversion can be related to any trait
that is a subcomponent of
agreeableness right because there are no
lines connecting them in that figure
but in fact the pattern of association
and covariation among more specific
personality traits below the big five
is considerably more complicated than
that would suggest
and so for example assertiveness is
negatively correlated with politeness
right so assertiveness is part of
extraversion politeness is part of
agreeableness
those are correlated with each other
however it happens that the

enthusiastic or sociable component of extraversion is positively correlated with the other aspect of agreeableness which is compassion and caring about other people and so you have two different components of extroversion correlating in opposite directions with two different components of agreeableness and those end up canceling each other out and so at the first at the other level you see no correlation between agreeableness and extroversion uh essentially so this whole thing you know the structure it looks simple in the picture but uh you know it's a lot more complicated uh the the lower down you go and so i would have you know i it seems to me that these aspect level distinctions are important uh they have they're they are supported by some behavioral genetic research as well as some of my own research just at the phenotypic level um i think that they're probably the most important distinction within each of the big five in terms of discriminant validity in other words you often get things that are predicted differently at that level um so for example uh politeness and compassion are the two aspects of agreeableness and well it turns out that uh if you

look at people's political orientation
people who identify as more left-wing or
liberal uh
score higher in compassion people who
identify as higher
in uh conservatism
uh more right-wing score higher in
politeness
right so conservatives are more polite
liberals are more compassionate
well what happens when you go up to
agreeableness you see no correlation
with political attitudes which people
found
surprising for a long time because it's
you know issues around altruism and
cooperation seem very relevant to
political
issues and policy well so it turns out
that that's because something lower than
the big five actually is associated with
uh with political attitudes in opposite
directions so i definitely think that
these lower levels of the hierarchy can
be useful
um you know there's less
the the evidence for the big five is
remarkably extensive right whereas the
evidence for that next
level down that i've worked on is much
less extensive
nonetheless you know again i think there
isn't just one way to carve this system
up because the patterns of
inter-relatedness are so complicated
and i do think that these are useful uh
useful
sub-dimensions at any rate even if you

could focus on others
so i'll i'll leave that there cool
thanks marco then scott
thanks um so the last few comments were
very very thoughtful
um so so let me know if i ramble on for
too long um
so so i love metaphors so if you don't
mind i'll try to provide a
complimentary one so instead of
maybe landscapes um i have this thing
with imagining
the brain as one big dynamical system
and so all those dynamical flows or
patterns they need to of course be
a pattern to order structured and
adaptive sense so
this kind of trick structure could maybe
be seen as
kind of a kind of grip kind of handle
kind of heuristic
um well of gravity to kind of
heuristically bias particular dynamics
and at least for me uh that would make
sense
because of these um big five factors
um being quite quite segregated in terms
of where i think they would be relevant
and that gives me back to the matter of
personality and and
kind of anti-moralization because i
agree that there's a big problem
of our theory laden in the manner in
which we approach psychology
um and that also uh
should exhibit itself in as mentions the
observations when it comes to animals um
but for example when it comes to

conscientiousness as adam already
very nice pointed out there's a matter
of death and i think
i think he did allude to it but another
thing to emphasize is also
um the kind of field or or a range of
possible alternatives to
to care about to someone who would be
low on contentiousness or carefree would
perhaps
uh not simply be lacking
goal-orientedness
but be more um aware of in touch
or have a bigger affinity with the
notion of play
of simply uh engaging in behavior that
is
reducing expected free energy
yet um not necessarily defined in terms
of a particular goal
right so not all animals would have
to deal with that right because the
evolutionaries simply
aren't trained or evolved to have uh uh
awareness or sensitivity to a multitude
of possibilities in what to do in
environments
and so the very emergence of a
particular big five factor might be also
related to
the complexity or the kind of yeah
the complexity of the imperatives that
they may encounter
in their life in their niches so
one may be interesting example is like
dogs or other social animals
uh although in general i don't think it
would be appropriate to

project or impose uh our entrepreneurial
notion of personality

i do think at least dog owners can agree
that dogs tend to have a personality
that seems to
develop or evolve and even seemingly be
similar to their owners

possibly because the imperatives of
their environment is expanded
by being in relation with human agents
and so by the expansion of possible
niches possible situations

um the relevance or the variation uh
specific to what consensus is
is a relevant factor uh becomes more
apparent whereas
uh previously that might not be the case
if that makes sense

um yeah i'll leave that that because i
forgot all everything else i wanted to
say

no it's really interesting that it's uh
big five is not the essence of an
individual it's actually
in a cybernetic framework making the
individual's phenotype in relationship
to the goals and the context of their
niche

so it actually moves away from some of
the essentialist types of critiques that
people might

naively bring up against big five we're
looking at this as an empirical pattern
with a generative underlying cybernetic
reality so scott

then adam than anyone else so
fantastic conversation the couple of
observations one

personality is for a person and so it's
kind of interesting when we take
personality and
impose it on other systems right i'm
just playing linguistics there a little
bit
but in a way we what is the delta
between personality and living
formality i wonder
so that might be in because are we
assuming the human elements again it's
that subjectivity question
one of the things also it occurred to me
that fristen it kind of
allows for a notion of situated
personality
beyond situated cognition and so maybe
those are
overlapping maybe they're not i'm not
sure if all um
personality uh elements are seen as
cognitive
or if there's some things that are you
know effective and hormonal and
whatever other causation chains there
are so many
it's interesting that the um the
behavioral elements
we ascribe the personality um
may or may not may not drive uh through
all living systems
i tend to define living systems and it's
not anything
reasoned it's just what i've done for
the last 10 years and it's been
effective for me is
i define living systems as auto
catalytic and

entropy secreting so they can be in a variety of different forms and so i think of markets and corporations as living systems and so to me one of the questions that with with the definitions in the fristen notion we have kind of a fractal notion of uh things that are show scale independence i guess is one of the elements of fractal systems and so to what extent can the definition of using these personality traits at different levels help us understand the scale and dependent possibilities there so that's one issue then the last point these are kind of random but i think they're clustered around the idea of the model ascribing the last one is you know when we talk if i'm talking to daniel there's at least four identities involved there my sense of me his sense of me his sense of him and my sense of him so one of the challenges when you're talking to corporations markets maybe not dogs because they may have quality is that the other um the other entity doesn't have qualia if you're talking to a corporation it does not have quality the people in the corporation have qualia but but the corporation itself does not and so the um well that's obviously we can talk about that

but my point is if they don't have
qualia can they have empathy
and if they don't have empathy maybe
there are differences in kind
up the fristen stack of fractal stack of
living things
and so maybe that is one way to start to
distinguish between markets and
corporations
and again gets back to the animals issue
do animals have qualia
is the quality of their quality is
slightly different than ours
such that maybe there's a subset of
personality that they'd
show things like that um anyway just a
couple of random thoughts i think but
they again that definition of living
beings i think is bound up in here
and again not being to us not being too
stubborn
in our perspective that the personality
is something that would describe all the
attributes that work up and down the
biological stack
thanks cool we got adam then stephen
marco colin
so go ahead adam
hi um hmm okay there's a lot uh
to try to write it in for me um
i like the so what marco was saying with
this um basically this this attractor
network formation
perspective the supposed metaphor
or a really good model the
he mentioned like niche construction and
he mentioned
um i i think this

perspective is really important in terms of thinking of the dynamic character of the processes that might give rise to these attractors over the course of development and situations and with development potentially being understandable in an open-ended sense that never stops um and so

i guess along those lines i'm thinking of that causal process model that colin has in the original paper where it's going from genes and environment to you're having these cybernetic function and then traits and characteristic adaptations

but then there's this feedback as and one way of understanding this feedback is as a kind of niche construction

relationship where you're modifying your environment you're specializing yourself for these environments

um as you're trying to like get along in the world as a goal seeking system

um one more nice thing

um with relation to what uh scott was saying i think that's actually

uh yeah so i don't know also like while i'm interested in seeing like how far we can stretch personality i wouldn't want to overstate it either because

there were major transitions in evolution and you know one of them would be

um basically the ability to have
certain types of generative modeling
where there might be a qualia
and quality of different varieties and
then we might go beyond that not just
qualia but things like
access consciousness and if you like
that distinction i do
like being able to know your experience
and report on it and things like being
autonomic and being able to
reflect on even your knowing um and so
this would be
very different cybernetic systems once
you introduce this
um so like you can have like and and
if you have something like the big five
showing up
or the the big ten or the medium ten i
don't know what you want to say but
um they might show up differently
so like you can think of like
generalized agreeableness as like
tendencies
towards like non-zero um interaction and
like not defecting in generalized game
theory something like that
like in a multi-agent context you're
more of a cooperator
and you you coordinate better but
very different thing if you're actually
having something like the feeling of
compassion or empathy um these are
different processes getting into the mix
and you want a different handling and
some things might not
actually you want a very different
handling

yep interesting like would rather
somebody act compassionately and be
perceived as high percentile compassion
or
perceived overwhelming experience of
compassion
stephen marco colin
yeah this brings in the idea of how
people relate in a task-based
environment or
a more task-orientated environment i
suppose where it might structure things
a bit more clearly
between the things in the world
where the person is and whatever
mechanism or structure
is interacting between the two i am i
picking up at all am i
there's some sort of delineation one
thing that sort of comes to mind
when we talk about so you've got the
generative model you've got these hidden
states of the world
and you've got this markov blanket and
the thing with the markov blanket is
unlike it's not a system boundary you
never can really define it
so you've got this and what i thought
was quite interesting really when
by having the stability and plasticity
i'm seeing sort of
the the plasticity with this
extraversion which you could say is like
influencing the hidden states in a way
you're perturbing out there
and the um the intellect is
making models is is the sort of the
other side

but the the the the um stability
in some ways is really stabilizing with
the markov blanket
like the the nature of the regime
because you've got of got
you know neuroticism is it like a is it
a very hidden
landscape of actually like inferring so
maybe we because i'm really interested
in how we might have mental space
psychology approaches which there might
be some sort of intervening
um spatial mediator to how we process
and that spatial mediator of how big and
scary things are around us and
in our metaphors it could be sort of
mediating so the neuroticism the
agreeableness to conscientiousness seem
to have
more structuring about how i might
take data in through a markov blanket to
the other side
and the other two seem to have a bit
more about what's out there
and what's in here in a in a very crude
way but
that's what i'm that's just something
that came up for me i thought was really
interesting
cool thank you marco collin blue thanks
um yeah i i also want to build up on
that so i fully agree
um so so i think i think the pro
the um prospects that are most
attractive with
this kind of approach to personality is
as stephen alluded to that that you can
uh associate these aspect factors

into particular imperatives to
particular challenges as it were
um and uh i think it's nice to know to
connect that with what was mentioned in
the beginning about mars levels of
analysis

so um mars levels of analysis is a bit
of a hot topic always

um so so forgive me i'm a bit biased but
personally i don't believe that active
influence should be seen as
quintessentially

computational because i think it would
discredit the fact that it implies a lot
about um the algorithmic level so

um just like uh steven kind of
already said you could have these
intermediate spaces right and that's
basically asking you how

these computational imperatives or
functions are going to be realized

um and i really want to emphasize the
notion of mediation because i want to
connect that to what scott was talking
about

so um it should be emphasized that from
what i can tell

this model of personality um is kind of
neutral about consciousness right it
doesn't we should be

very very um mindful about not
conflating

the first person signs or
perceptual models of how empathy arises
within us

with the process of empathy as such
because

if you see the pros of empathy within

the brain then there's two ways to
perceive it at the very least third
person view
from measures whether physiological or
questionnaires
and first person of course um and it
stands to reason that that
it's not necessary to have that first
person encounter with these processes
for this process to happen but having
said that
it would be very interesting to to to
imagine that consciousness would be
like a sp intermediate in space as well
such that
the component factors and the respective
challenges
of the big five um can actually
equilibrate in the sense of
that it's not a trivial challenge to
balance the respective aspects of these
factors
and the factors between each other for
the uh
autonomous being of concern and so it
would stand to reason that indeed
um there might be another intermediate
space intermediate between these five
factors
which would emphasize the cybernetic
function of something like
consciousness as a space which i think
it might be interesting for you scott
because i think then you can generalize
it
for non-living beings or or composite
organizations because if you then
associate personality as heuristics

for certain domains of challenges or
domains of imperatives
and then uh having to bind them together
through a certain integrative space
then you could have i think a very
interesting approach to organizational
science if that makes sense thanks
that reminds me a lot of the projective
consciousness model paper that we read
in a previous active stream but it's
almost like wondering if the cybernetic
organisms or cybernetic systems other
than organisms if they have a projective
geometry

do they also have that awareness so
colin blue

scott

what i wanted to say actually ties in
nicely to the point that marco was just
making i was going to start by saying
that i think it's important that
the existence of personality is not
dependent on

the existence of qualia so

this is a i think a mistake that people
often make when they're considering
the big five and similar personality
systems they think oh well these are
just about how people perceive
themselves or how about how people
perceive

other people and maybe they're useful
categories in terms of like what you
need to know about a person
but actually the claim is that you know
assuming that these descriptions of
people's behavior have validity
they describe patterns of behavior that

exist regardless of whether people
are accurately aware of them or not and
so you could have
differences in you know persistent
differences in behavior from organism to
organism
in any species that would be directly
analogous to what we think of
as personality in human beings now of
course it is true that we have this
additional layer in which
one uh kind of characteristic adaptation
that we have is that we tell a story
about ourselves consciously
and we can describe ourselves and we
actually make use of that when we try to
measures
measure people's personalities because
we you know often just ask them to
describe themselves
but we're not interested just in when in
their description of themselves as such
we're interested in or we hope and you
know we know from a fair amount of
evidence
that those descriptions are reasonably
accurate so it's not just that i think
that i talk a lot i actually do talk a
lot if you
if you take some kind of objective
measurement of that right that's part of
what being extroverted is about
um and so you know consciousness is
obviously
something that well is mysterious in
various ways but i think that you know
if we
focus on that division between the easy

and hard problems of consciousness i think we actually have some pretty important things to say about the so-called easy problems of consciousness like

how does it function what does it do um and you know i think marco is right that one of the things that consciousness does is allow

people to deal with conflicting information conflicting goals and to to create a sufficiently broad model of the world and our attention of it that we can then control

our attention effectively i really like jeffrey gray's

uh perspective on consciousness his last book was just called consciousness creeping up on the hard problem um that's really excellent and then uh i also like uh what is his name michael graziano's

theory of consciousness it ties in very nicely to this idea that a cybernetic system

uh to be a good cybernetic system you have to have some kind of physical representation of the variables that are being controlled that's the good regulator theorem

um and so i think that one useful way to think about consciousness consciousnesses as

a model of our attentional field that allows us to

uh direct our attention effectively to things that are important for our goals or things that are potentially

conflicting between goals
and so in that sense yes it would
certainly be relevant to
conflicts that arise because of
different needs or goals that are
related to different personality traits
but it's not something external to the
traits mechanistically because the
traits
are descriptions of patterns of
variation that you know
in a population and so when we ascribe
them to the person the person has a
level of the trait
and so those traits actually should be
reasonably comprehensive descriptors of
functioning in all the important you
know categories of human function and
that includes consciousness
and in fact if you look at the contents
of the big five openness and intellect
is the trait that is really
uh related to the qualities of
consciousness
people who are high in openness have are
conscious of more things their
consciousness
is broader it's deeper it's more complex
um
and you know so there's the openness
side of that trait in the intellect side
and so you've got this sort of pattern
detection
and then pattern analysis and all that
is basically about
uh in large part the qualities of our
consciousness
cool very interesting blue then scott

then adam
so i am going to totally bypass
consciousness because it's just
uh it's a hard problem right but um i do
want to
to just consider for a minute what would
happen if we
maybe maps like not not asking what is a
person but what is an individual
could we then like scale over like
non-human systems
like you know organizations or even to
other
animals like adam brought up like major
transitions in evolution and how do
these like personality
traits maybe influence evolution that
would be interesting to think about like
can we think of maybe clams as like
ultra neurotic
features and ants as like high in
conscientiousness
could we take like these personality
traits and map them onto
organisms you know that have evolved
over time and and what would that look
like in terms of
of evolution and what what is essential
you know openness to exploration is
obviously key
for you know some of the major
transitions to happen
that reminds me of the principal
components the big fives um
they're based upon the patterns of
empirical variation in the actual
population as
colin brought up and then the animal

spirits are
without making a metaphysical claim
they're like archetypes like of bravery
or of extraversion or
social butterfly or you know a couch
potato it's still a living creature it's
a potato it's in the ground
and so how we model different types of
metaphors it's like that's on our
procrustean bed but then what does it
look like to have it on their
procrusty in bed for their personalities
or individualities
so scott then um
marco then colin blue i love the notion
of neurotic clams
that's my favorite
um yeah this is this is great and just
along the same lines a couple of things
that leaped into my head i was thinking
of irving goffman stuff where you
project yourself out as you want to be
perceived
so that's kind of you're modifying your
environment in terms of the persona
perception
um that was one thing then also there's
a lot of self other things in here the
identity kind of notion and that and
that idea of presentation of self
one of the things i was wondering about
is the dsm diagnostic statistical manual
dsm-5 just came out
and really how it plays into the um
what i guess they would define as
anomalous persona
traits um and usually dsm seems to be
the ones that aren't

the ones that make you not productive in
the capitalist society kind of things
because there's a lot of things in there
you call cultural in there but it's
interesting
that those are the outer bounds of uh
what's expected expectations of persona
and i think a lot of the survivalness
questions have to do with expectations
and the uh
the rising of these variables in the
corollaries in animals and in people
have to do with survival and
increasingly social survival but also
just that survival of the organism
and so that idea of presentation of self
uh might be even related to camouflage
and issues of that you know how do you
present yourself or not present yourself
in society
so anyway that just that host of
thoughts that how that naturally evolved
is kind of interesting also this is and
this is not a a
studied thing i'm not an expert on this
but just functionally again
i think of the um mind
and consciousness well i think put
consciousness aside just like blue said
but i think of the mind as not in the
brain i think the mind as in language
and material culture and the brain as an
antenna
tuned into the local mind so if you're
my
sister adopted a child at the age of
five months and now she's a
was raised in central pennsylvania and

she was adopted from china she doesn't
speak chinese she speaks spanish in
english she's a u.s
person but she looks like a chinese
person so that their mind was developed
in
in in the in pennsylvania and so
if there's no feral consciousness which
let's take that as an
assumption for a second with no feral
mind and if it's a learned phenomenon
it's interesting that we have this
framing and then we
keep re-upping the framing reaffirming
and valorizing the framing
it's interesting to wonder if there's a
lag
in terms of survivability what were the
persona
characteristics that were selected for
and i don't just mean
in biological evolution but socially are
they identical to the ones that will be
selected for in a much more
dense interaction environment for
instance so there may be some
interesting questions
if and to the extent that persona is not
on board in the brain inherently but is
a
learned social phenomenon what is the
how malleable is it and i mean i guess
that's what education is about and why
colonialists try to get indigenous
people in their schools
their children in schools early to
render the externality innocuous
but it'd be interesting from a

perspective of
current situations that we have in
current policy challenges and public
health challenges etc
information challenges how might we
handle those
recognizing a more plastic uh
persona so just a few thoughts
cool interesting marco colin adam
thanks um wait i forgot what i was gonna
we'll come back
okay well i think that there are
certainly uh i mean certainly true that
there are aspects of the mind and
personality that are learned but then
uh there are others that are
evolved essentially and sufficiently
genetically programmed that they would
be present even in
you know feral children but
and that connection back to the idea of
the evolution of personality uh when i
when
you know blue is talking about neurotic
clams i was thinking about some of the
work that's actually been done
on the evolution of personality and
animals
some of it is very interesting there's a
really nice paper from
oh i know almost 15 years ago now by
daniel nettle that talks about the
evolution of
personality and humans and other animals
and so one of the things that he talks
about are these experiments
uh where you take uh a bunch of fish
and you if you just

naturalistically observe them you will see that there are differences in personality some of the fish are relatively bold some of them are relatively timid and cautious and what you can see is that if you systematically manipulate their environment you can actually change uh evolution of that population over time in interaction with the personalities of the individual and so the way that i think about this is that personalities drive evolution but evolution also drives changes in personalities at the population level so what happens is that if you have a if you have those fish in an environment that has a lot of predators then the bold ones get eaten and the frequency of the more cautious timid ones increases in the population but if you put them in an environment that has very few predators then what happens is that the bold ones uh tend to monopolize a lot more resources and the frequency of the bold personalities increases in the population yes so always the ecological selection on the development of personality not even just personality something that can be reached out and grasped but actually how personality becomes developed in relation so adam then marco then stephen try to tie together a few threads before

i lose him um
so like earlier i was saying
um you know evolution is one of multiple
perspectives but
a really special perspective and it is a
really special perspective
like i it almost makes me move towards
like well on some level i'm uh
an instrumentalist um but with respect
to like the things like
what is the ontological status of the
markov blanket
i think there is also like a realism
like
realism and maybe a realism so strong we
couldn't sometimes throw away with the
word quasi
we should always kind of keep it in our
back pockets and that we should keep
keep in mind that there are multiple
ways of carving up the world actually
infinite ways it's just some are going
to be more or less good for being such
as us and
one really good carving would be the one
that basically
relates to functional cycles that
achieve closure on time scales where
they can be subject to
cumulative evolution broadly construed
whether being the development
of the organism or phylogeny and so
uh what that might uh
pick out some particularly uh
universally agreed upon while still
subjective joints
but to kind of bring it together more
among those

and with relation to your personality characteristics there's an interesting thing that can happen for species capable of modeling each other and themselves with a certain kind of richness having certain types of consciousness in the sense of knowledge

where um and even without it potentially that they can become aware of these reliable

course screenings of themselves and others and then

this forms a basis for your planning

and then this forms a basis for

basically not just the niches that

you're trying to construct but the way

your niche is constructing you

and so for all this like active

inference with its like embrace

an embracing of circular causality and

this mutuality between like system and

environment in terms of

each being its own kind of active

inferential system and exploring this

interplay

i think that's really essential for

understanding personality in both the

specific

and the general case uh before i'll

let you all move on like one final uh

set of threads

i would be like when talking about

something like consciousness so like

i recently kind of took on the hard

problem uh

it has debated how much success i had

but in trying to make way on this

um it seems useful to
not uh and probably with respect to
discussions of personality
um all of our words are just words
they're uh course screenings that we
agree upon um
and they don't exist in the world and so
consciousness part of the
issue is like we mean very different
things when we use the word sometimes
and so um so similar so you might say
um scott was suggesting like you know is
um
we might want to distinguish between
mind and consciousness or intelligence
in consciousness
and so you could say like your mind is
distributed
through the world and that your brain is
an antenna
i think that's a good perspective but if
you're thinking of like
consciousness as like a type of model
where there's something that it's like
from the inside to be that
kind of modeling process that might not
be extended
that might be like a subsystem and a sub
model um
and so it's similar we're kind of
building up this onion of complexity
with respect to consciousness and people
are kind of kind of
engaging at different levels but not
always being clear on the level that
they're engaging it
and so you could so recently um i was
engaging at the level of like

basic phenomenality itself um and
uh rudroff and like projected
consciousness models very similar spirit
um it's uh you know specifically uh
what are the ways we can describe your
subjectivity as a particular reference
frame and point of view in the world
as you as a cybernetic system how can we
characterize that
and so then we go like another level
still we're getting what colin's talking
about with things like
attention schemas and like
models of your awareness for for
modeling yourself and others and shared
uh and shared attention schemas and so i
guess the
go further still into things like
self-consciousness reflexive
self-consciousness
and each of these will be kinds of major
transitions
in evolution and development which i
understand as a kind of evolution
and each of these will
now dock with personality in different
ways
um now i guess the one final thing with
um
that's interesting uh is the way
in which personality traits if you're
thinking them as kind of
coarse screenings it's like with like
coarse grain descriptions
and it's a one key core screen you can
core screen when you have something like
a renormalization process when you have
some sort of like emergent attractor

structure

um such that the lower level or the
smaller fluctuations

uh smaller dynamics can be kind of like
averaged out

because now you're pointing at this
bigger thing and the bigger slower
attracting manifold didn't care about
the particulars

so uh rather that's not relevant for
this

course or phenomena that you're trying
to observe and so

there's a sense in which um
personality is a kind of correspondence
like what are the nature of these like
normalization processes and so one of
them

it would seem like so what is inducing
this attracting structure

it seems like a good part of it would be
so you can model yourself
across situations so you know what kind
of agent am i

that i am likely to be doing what and
what circumstances

and this will be pretty essential for
pursuing any kind of complex plan
but similarly for coordination um other
or

working around other agents um or
cooperation or

competition what kind of agent is this
i'm dealing with

but for the sake of cooperation you
might you know ask someone

to be uh uh responsible

they should be able to respond and

report

in in a reliable and consistent way

we actually demand of each other in a way

a certain non-erratic character we we demand dimensionality reduction of each other

um just because otherwise we get surprised and as a friendly principal and active infrastructures we don't like being surprised

um because then we can't plan and then we can't outsource our cybernetic entropy

and it's it's bad so there's a sense in which um

to what degree like there's this the same way that like personality traits are influencing they're emerging from evolution then influencing the course of evolution

with development uh your personality traits are emerging from your interaction

with your environment as you're carving out this niche um

but also they can also become like attracting targets and of themselves on different levels like if you're uh if you have this expectation of yourself

as being this type of agent it could be a self-fulfilling prophecy

i think i'm a neurotic agent i think i'm an extrovert agent

now other people think i'm neurotic agents other people think i'm an extroverted agent

and now they're treating me a certain way and i'm acting a certain way and this

and then it keeps going um yes a lot of things you know a lot of no a lot of awesome stuff that reminds me of the scripts

it reminds me of the phrase uh the common saying the absolute power to act absolutely brings responsibility or something like that like power and responsibility but

now we're talking about the power to act and the responsibility to stimuli

and that means that we respond so um marco

then stephen thanks

um adam for that fantastic

um so first um a bit of a meta comment or question so since you said we can't use the chat in stream

um i've been putting some links in the youtube chat but i'm wondering if you had a recommended way to do that because a lot of people are mentioning a lot of interesting research

and i think people are modest to the point that they won't advertise their own research or link it

yeah um so what would you prefer daniel sure

to respond yeah to respond to the meta and just to clarify so

in the jitsi for the live stream it will make a

if you use the chat so it's very disruptive to use the chat during the jitsi

so the single source of truth for the
video is the youtube page
or on keybase so definitely everybody
should make a keybase username and join
us at

actinflab.public and check out
activeactiveinference.org to find out
more but

the key base at this point is our actual
organizational hub

so that's where we're doing the open
participatory active inference lab

like this is a project of the
communications branch of active
inference lab

continue okay thank you yeah i didn't
know but key basis but thank you glad i
asked

um yeah so building on saffron's uh
great comments

um so i fully agree that personality is
definitely something scaffolded right

um so um personally i

i i i do see this at the the relevance
of of course spreading but i think it's
also um

maybe interesting to use another uh
example or metaphor

of bootstrapping so um

one interesting thing is so an ai keep
dealing with interactability but clearly
we are still

living and not defying laws of
computation so

um there needs to be this kind of leak
of faith of how to start off

acting around in the world uh and so so
so

one way to see it instead of terms of
course grading in terms of temporal
scale

maybe it's simply um then put it this
way

it's going to behave over time over time
over time across different contexts
there should be overlap in the kind of
parameters or hyperparameters
that govern your behavior approach to
the world and

those that are more consistent or more
covariance or synergetic

they will consolidate into this kind of
heuristic

core to further elaborate and specify
um and on top of that building answer
that i was wondering because adam
earlier asked colin if you could maybe
uh

explicitly map the uh the big five to
mechanisms

cybernetic or inferential terms but i
was wondering if it's

um appreciated or useful to also put the
question

for everyone since everyone's relatively
familiar with active inference or
fustonia terminology

how they would characterize um the big
five factors because mostly it's

in very simplistic uh

adjectives or purely behavioral but i
think would be interesting to maybe
at least attempt faithfully to

understand them from the inside out
right what kind of functions or
processes or inferential problems

would be associated with them a few
people already did that but i
was wondering if people would like to uh
do that more
explicitly uh together cool
great question and also a lot of it is
enclosed in the paper which is why this
is so fun to have as a journal club
because we can always
superfluously add on top but then the
paper also references a lot of these
things
so we'll go to steven colin scott
yeah a lot of really cool stuff um i
think
um well as well as the modeling idea i
think going back to this
the the the idea of a timid fish and a
bald fish and
the affordances of the environment one
thing that's really been
i've been really striking me as has been
trying to integrate
um drama-based projects and sort of
experiential workshops with
psychology is is that
you've got this kind of reflective
perspectival
aspect to psychology like you can take a
perspective and reflect and you get a
sense of it
the big five will use that in the
reporting um yeah it's tapping into
some sub-personal dynamics and other
dynamics which may not be
available at the conscious reflective
level
and i think that that that then starts

to get really interesting with the
affordances again with the timid and
bold
and what's been talked of of how this
all sort of scales because
um it's not necessarily
a timid generative model that creates
the timing like
if i take too many cups of coffee okay
now my i don't know if it's my brain
that's changed or whether it's the
sensitivity to my
um of my of my
perceptual and active action reflexes
which are kind of now or maybe my
stomach or
so you've got this um there's an
interesting way with this that it can
bring in
the kind of the dynamics and the
affordances and
so one thing that's quite interesting in
in this in theater you've got an acting
method which is basically
very like you hear a very text
orientated and what did my character
want to do but another way is you just
actually get into the body of an animal
so you be a spider
and they actually have in the
school they do like a set a whole series
of weeks in the zoo
and everyone has to be animals and
inhabit animals bodies
and but you know there's something
different about if you're on four
legs and you're moving around smelling
things okay

you're going to be balancing your inputs
and your options and affordances for
action
are quite different right so it might
well change the way
um but it will change well i'm not going
to speculate on that but just
just basically coming back to the point
of the timid and the bold and
this idea that the the way the active
inference or the inactive
sort of dynamics of the body with the
environment
can be a lot of that could be sitting at
a sub-personal level
like you're not really sure how you know
that bright light has affected my
retinol
or my tiredness has affected or my
alcohol consumption
you know some of these types of things
so i thought i'd sort of say that that
kind of
can connect in with this the more
consciousness
reflective piece and the sort of
phenomenological
cybernetic subconscious piece
cool colin scott adam
i just wanted to say that the reason
that i didn't take up
adam's suggestion to go through each of
the big five and talk about the
cybernetic processes that we think are
probably related to them is precisely
that they
you know that is all discussed in the
paper i didn't want to take up a bunch

of extra time of things that
um many people have already read um
in terms of this kind of idea of the you
know the richness of the psychological
processes that are underlying
each of the big five that is something
that
you know it's true that a lot of people
have this very shallow perspective on
them
um partly due to the way that they were
derived in the first place and those
stereotypes that people have about them
as
you know scientific entities but i could
spend
the rest of the livestream talking about
these
processes in great detail but i think
would probably be better just to refer
people to the original paper that i
wrote in 2015 on cybernetic big five
theory
and then also maybe to a theory that we
wrote on the cybernetic perspective on
psychopathology
because the big five turn out to be
equivalent also to the major dimensions
of risk for different forms of
psychopathology and we talked a lot
about the specifics that might be
important there as well and that all all
my papers or most of my papers are
available on my website
cool great and that's also why again the
journal club format it's not the only
format we're going to experiment with
in active inference lab but this will

actually allow us to
have conversations that are rooted in a
lot of specifics
and various papers will be amenable to
that in different ways

so scott and then adam
so and i wanted to apologize for not
reading not having an opportunity to
read the paper before this one so it
would have been better informed had i
done that

one i wanted to um this just occurred to
me

something that might be going on here
and it's a little bit stretched
so let me give me a second here so this
is going to be where i'm going with this
is whether living forms are entropy
engines okay that's where i'm going to
go with this

so um there was uh

[Music]

entropy i'm talking about now using
shannon and von neumann
and um entropy notions generally
not by analogy but rather saying you're
talking about general disorder and
and so i'll be a little fast and loose
with entropy for a second here
so penrose wrote a paper a book recently
we talked about the what we get from the
sun is not
energy but rather neg entropy because
the yellow light of the sun comes in
infrared light goes out the difference
is the frequency the information
carrying capacity of yellow light is
higher

because higher frequency and so it's a
net energy
neutral except for global warming or
climate change
but it's um what we get is the structure
of plants reflects that
higher neg entropy opportunity i guess
is one way to put it
so let's so starting with that um what
if entropy was driving a lot more
and so what if the um
remember recall my definition of living
forms as auto catalytic
entropy secreting entities
so if entropy universally is increasing
living forms are localized neg entropy
areas right they lower entropy locally
and try to
preserve it what if the the drive
both the physical symbiogenesis
where pre-existing bacteria migrate
on board to eukaryotes and then become
mitochondria
in human in animals and
chloroplasts in plants so they come
together because it's a better deal
basically a better way to be auto
catalytic and entropy secreting at the
larger scales
what if that extends um to where we are
now with the
with the the what if those stage stages
of perce
personality have those functions we're
saying they do
and um then it has to do with both
entropy accounting
so when you're meeting someone who

shares personality states or has
predictable and ex
personality states you can predict
there's a lower um
differential so from a second law of
thermodynamics perspective
you have like for carnot's equation you
need a hot and cold differential to
perform
work what if the information
differentials
that you have between entities that are
meeting each other
help to cultivate various responses
and those responses are we are
cataloging in those
five big persona traits and so that's
ways of being or ways of encountering
um so it helps you navigate the social
landscape
and um that the what what if what we
have really is what i call an entropy
engine
which is that the personality goes out
in the
and harvests disorder um
through these mechanisms dissipating it
obviously but also
um growing from it and what i mean by
that is
all institutions all of them are
uh dis de-risk some subset of
interactions whether it's grocery stores
or hospitals or banks or whatever
that's what institutions do at larger
scale so they're neg entropy expressions
at larger scales
but we have those um those those feelers

that go out
and maybe the personality types and our
physical attributes
are the feelers that go out to figure
out where there's opportunities
for proliferation into a community of
interest around a shared risk
thanks wow very interesting with the
jordan peterson the darkness into light
ordering the universe and you know or
disordered order as a thermodynamic
inference
process so hopefully that will be
something we can like continue thinking
about in our last
20 minutes so we'll do adam then uh
marco
and anybody who's watching live can
leave a comment anyone else who's on
here can prepare any
like last thoughts but this has been a
super awesome discussion so adam then
marco
um with relation to entropy
um you might want to check out the work
of uh
in terms of life jeremy england he has a
really interesting
handling of that's free energy principle
compatible
of entropy maximization and you might
also
enjoy terence deakins work colin
hooked me up with him it's incomplete
nature he gets into
um discussions like cough uh
auto cut discussions of auto catalysis
like kaufman talks about

and um uh work cycles as ways of like
describing systems and their reality
because
you know if you're talking about
realities you know differences that make
a difference uh
being able to contribute to a work cycle
is a pretty good criteria
and if we're able to identify system on
some sense it's engaging probably
in some kind of work cycle and you can
do an entrap like a handling of it
that's going to involve entropy
uh also in terms of entropy um
there is maybe two papers you might like
one is um
i posted them in the youtube uh feed
which is um
uh psychological entropy um from
uh hirsch mar and peterson 2012
and then a more recent attitudinal
entropy framework
um by daleigh bores boom harvard and
van der masson forget my pronunciation
uh but to
now look back around to some of the
things um that
stefan was saying uh that still on some
level abstraction will involve entropy
um one of the places i'm actually really
thinking uh
now into and trying to understand is uh
the conditions under which personality
might be more or less plastic
the conditions under which you can get
trait change altered traits
um and not just altered states and uh
along these lines of notions of like

acting like technical acting versus
method acting
like to what extent is this sort of
self-fulfilling like the
self-perpetuating
attractor dynamic that makes up persons
and their personalities
um to what extent
uh is it changeable even if there are
these enduring cybernetic
parameters to what extent can you take
those arrows and columns causal diagram
and really work them so collins um
basically
talked about like if you want to have in
his well-being work in his um
uh psychopathology work um about
character adaptations
being where you have the most leverage
like the things you tend to do
the this like design of your life and
the situations
you show up in and how you show up and
this is where you
like or clinicians and yourself you
might have the greatest purchase
in an acting change um but
and so this would work on multiple
levels one would be like it would change
the nature
of what you experience and you know
we're learners we update
it would also change your niche
construction and change the way that the
the context in which you're in
if you if you that would be leveraged
there
but um i guess to one more

thought would be um and bring it back to
consciousness

um like to what extent

it's it's not just chance so to change

your characteristic adaptations

would be to change the things you tend

to do um but what you're describing this

sort of like method

acting view or are you bringing that up

but like acting

um what extent can personality be

changed by

really paying close attention to this

what

is it like to be me and actually

and what could it be like to be a

different kind of me

and then imagine yourself

in that way um i won't quite call it

faking it until you make it but like

um and to what degree

is this a substantial part of how

personality forms to begin with like

children at play copying their parents

pretending to be

this fair that way but now the question

would happen in the bootstrapping

can we still leverage this um as adults

and one thing along those lines would be

um

that makes me think you might be able to

um i believe in the

uh tibetan tradition though um i think

it's part of jokes

i don't know how to pronounce it but

basically um they'll do these kinds of

like uh deity practices where like let's

say you have some kind of like

affliction something that's like
not working for you um you might pretend
to be a kind of complementary
uh deity or spirit but
you don't pull this off on your own like
the social context
and so you create this kind of sub
persona that you live into
i believe they call it tulpa marco i
think actually pointed me to this
um but uh but you don't do it on your
own you have a holding in hardness
social context
that supports your act and you're able
to kind of collaboratively
enacted and that seems to me to be agree
this is
we can you know empirically establish
this as a mechanism personality change
this seems to be part of what makes the
magic happen so
uh yeah that's the things i'm sort of
thinking into like what are all of the
levers we have
to actually design our personalities
going forward
so you know what mental techniques uh
psychedelics like meditation what what's
at our disposal
so that we can have like you know we can
be
good meta regulators over ourselves and
like
realizing like our deepest values thank
you
adam i'm just gonna throw up that
closing slide and then we'll
hear from everybody who wants to raise

their hand with the final
note um this has really been an awesome
active stream thirteen point one and 2.
so in the calendar event for those who
are participating live
there's a feedback form a survey which
would be really appreciated
and all of you are always welcome
whether you're the author or not
to participate in a future active stream
you can go to the spreadsheet and find
out the calendar
but all of you would be extremely
welcome to be in a group active stream
like this again
and then if anybody has an idea for an
event that they want to arrange or like
an interview
or a follow-up discussion or a few
minutes isn't enough during this
conversation to talk about drama
or consciousness or whatever it happens
to be let's just plan that as an event
so that we can always
make sure to hold the space for hearing
about everyone's perspective because
it's just such a great
stew when we can have it all going so uh
marco and then stephen and then anyone
else who wants to raise their hand on a
last
uh note thank you
i agree this has been very very
enjoyable so thank you danny
um yeah scott thank you for the nice um
metaphorical pictures so i fully agree
that there's this beautiful notion of
the feelers right but

but i think it's also interesting to um
first
distinguish reflexive feelers and
intentional or expressive feelers right
so how do you navigate
uh to the world and in that challenge
how do you navigate yourself and so i
think personality or yourself model
regarding you as a person are more of
those reflexive theaters
and then subsequently you are more able
to grip or control or
regulate the way you navigate the worlds
with your perceptual feelers
more specifically the the recognition
models by which we are able to take in
the worlds
because they are what are being compared
with the sense data um
and you also said something very nice
about i i really like
carnot engines as a picture i actually
love that you mentioned that
um and you mentioned information
differentials
now information differentials uh is
exactly what active inference is about
right so so so the beauty is also
there's a notion
you mentioned entropy a lot but there's
also a notion of relative entropy
which is also known as the cool block
library version which is exactly
what underpins the entire formalism um
because exactly like the carnot engine
you want to minimize the temperature
differential but in rain
temperature is constant right similarly

the energy consumption doesn't vary with more than order of magnitude and so the interesting interesting question here is okay to what extent is it like a carnot engine

to what extent um does information just the minimization of information differentials play a part in this functional behavior so one very nice notion that i like which i got from a friend said burnett shout out um is coordinated equilibration so both in the case of normal conventional thermodynamic equilibration such as incarnate engines as well as in my opinion for active inference you're basically equilibrating the whole collection the complex covariance and co-conditional collection of information differentials and so no matter what's by virtue of it being physically instantiated it needs to equilibrate physically which has some relation with informational equilibration so in other words one thing i'd like to say is basically we're letting nature do the work right we're already nature it has to equilibrate and it happens that that finding of equilibria is in itself homologous or analogous to particular algorithmic instantiations which if you're interested is also interesting in computer science because apparently there's some nice works that connect um icing spin models

at three models of magnets with a
self-organized criticality
right how the brain operates via
avalanches or cascades of
activity apparently these are formally
implementable to solve basically
most famous algorithmic challenges so i
think hopefully you might find that
interesting um right so
yeah and so i'm not sure what your exact
phrasing was but i want you to note that
i think it's also a nice picture to
imagine that what we're doing
effectively
oh yeah yeah you're talking about neg
entropy so in fact what you're doing is
basically
you're you're partially deciding how
you're going to be imprinted by the
world
right and that's the whole power of the
whole predictive picture
because the patterns that are out there
in the world um are
effectively the challenges themselves so
when we internalize the patterns
then we're more gripped or attuned to
our or have a bigger affinity
with the world so exactly right uh in
that regard
uh yeah sorry i forgot the rest of what
you said
um and regards for adam's beautiful
point about spasticity of
a personality um i also very much echoed
this
kind of criticism slash question of
the whole assumption that personality

traits

is static right it's rigid but i think

what he said earlier about course

grading and what i called bootstrapping

kind of maybe is a path to understand

better how it

shouldn't be seen as so static because

if it indeed is

a bootstrap in a time where it lacks

self models and basically the

accessibility of changing that initial

bootstrapping model

is uh is is inhibited or it's obstructed

effectively right

because if you want to reflexively act

upon these models

then you need to have a path towards

these models

so that you have to um act upon whatever

influence your personality

through layer and layer layer from

superficial to deeper right um

and that's a big challenge and i think

indeed things like topaz things like

visualization exercises

are one example but i also really like

siemens notes on

drama and more situated experiments

situated design

where i think the notion of porosities

was very interesting right to what

extent do you allow the environment

to influence you um and finally i just

wanted to note

that's because i mentioned it uh

psychedelics yeah absolutely because the

whole

phrase is psychedelic the revealing of

the mind

but that's only one step the second step
is reveal such that it becomes
accessible to to systematic change um
so yeah so so those are my thoughts and
thank you for

giving me these thoughts nice it's
really great how you tied so many of
those elements together
so in the last couple of minutes just
quickly from anyone who wants to speak
we have stephen
then scott then anyone else who raised
than adam than anyone who raises their
hand

i mean this is what this paper is so
interesting about is the way that it it
gives these this dynamic mechanism
which is good for active inference but
it's also

also quite an intuitive way to break
down all these genetic models and
and different things it gets hard when
you start to layer different areas i
suppose the two things that come up when
i was thinking about
the comments on the drama or acting is
um

is i think sometimes the acting or
having a chance to go and do
a performative event or action that's
more like you're

you're going on stage and
that could be like that mounting car
problem i don't know if you see
if you see the paper of exploring a
landscape

um using your model and seeing different

possibilities so it could extend
out the possibilities and that mounting
car that
problem was quite good for looking at
how that issue of expanding out
can be important and then there's also
and there's different ways that for
instance method action is actually
can be quite dangerous a lot of people
have psychological issues if they go
down that route
um so the result there's a basically
putting on an act
is like you're still a performer and you
know that you're in a sort of a field of
practice
presenting and being aware of where the
audience is how close they are what size
they
of you they can see however
just by doing that you will have that
experience so you will have expended
your
expanded your range your entropy of
possibilities or states of the world
there's also participatory type drama
where you could revisit
a real life experience and go back into
um through psychodrama social drama an
actual experience that you had
and look at that and then that also
could give different ways to
maybe understand your actual lived life
now whether that would change
the big five i don't know it may not be
about changing the big five
for someone it may be something else
within that you know but it um so i

think it's really
there's a i just thought i'd add that to
the great competition
okay thank you scott and then adam then
colin just in the last couple of minutes
this is great very rich discussion i
like the fact that we raised more things
and this is really the beginning of a
discussion
than we resolved and that's beautiful um
the tilpa really struck me because it's
something where there's an imagination
in an internal which is re valorized
reified by the external in a
in a process way and it really reminded
me of law
in law if you say you have a right but
there is no duty
of anyone else to respect that right
like if i say i have a right to free
speech a right to sit anywhere i want on
the bus
and no one else respects it i don't
really just i just have words on paper i
really need the duties and the rights
and that dyad i think is really
interesting that reification that
valorization
and it started to wonder whether it's it
feels like something like behavioral
catalysts or consciousness catalysts
where you have that external and
internal element
are necessary for that coordinated
equilibrium
equilibration that as marco was alluding
to so i'm fascinated by the discussion
here i feel like there's we're just

beginning

and looking forward to other ways for us

to continue because there's a lot of

pieces that we still need to explore it

feels like

totally agreed uh so we'll have uh

adam then colin and if anyone has

thoughts on where we could explore next

or what would be fun to explore they can

mention it too adam then colin and a

special thanks to both of you authors

for coming for

both of these conversations it was just

it really made the whole thing come

together so adam colin blue

yeah i've been loving this thank you all

of you

um i'm really glad marco brought up

criticality because

it gives me an opportunity to hold forth

for uh

that's what colin knows is my religion

um

the and i think this someone harkens

back to last week's discussion of um

meditation and mindfulness um so one of

the ways um

i view mindfulness is a kind of like

psychological manifestation of

self-organized criticality

so it's like there's this edge of chaos

or this enter regime

where you have enough uh variation

enough disorder

to give birth to a dancing star or

rather to have variation for evolution

to work with but then you have enough

stability to get this cumulative

evolution to have structure built upon
structure
and so on and so if it's the one regime
where evolution is possible
yeah this is a multi-scale phenomena
this is probably this is a universality
class like you'll find it everywhere
at all scales and so at the
psychological level
um whatever you're really doing
um in some sense i think it's going to
be involved
with respect to optimality and if one of
the things you're doing is trying to
like
uh for instance uh try out something new
um and like regulate like the amount of
psychological entropy you're
experiencing
um this um basically setting up for
yourself this return to this metastable
place
where you can balance stability and
plasticity where you can sort of
ride along the edge of chaos and not
fall in or
or go too straight too far away from it
um this is
i think if you want to um either
maintain your personality or change your
personality
or work within the bounds of it
this seems to be really essential and uh
i i
this final thing is like i've been
increasingly thinking like the way we
think of mindfulness
is like all wrong and that rather than

thinking of it as like
um this is a really nice state to be in
and you're watching your breath and
everything
is like super great that's good but like
more like the face space description
and more as process and rather than like
a thing you get
it's a return to like a kind of
a unnecessary home and really an
essential part of cybernetic functioning
of balancing stability and plasticity
that you're going to need for anything
thanks so much adam colin then blue
i'm glad this whole idea of the edge of
chaos came up because
adam earlier you said at some point you
know we're talking about active
inference so
talking about free energy and we and we
don't like surprise
and of course as soon as i hear that you
know what i think well no we do like
surprise
um uh we seek out surprise often we like
surprises right we like
uncertainty we like uh we like risk
people enjoy these things that's
exploration
and so and the reason that we have both
it's
one i think maybe the most important
fact in psychology that we have an
innate
ambivalent response to uncertainty and
an increase in entropy which is
simultaneously positive and negative
right we

obviously don't like surprise we fear
entropy we need things to be predictable
or everything disintegrates
but at the same time we like surprise we
like uncertainty we like exploration and
if we don't do that then we don't adapt
and eventually we get too rigid and
everything disintegrates right so
you have to seek out entropy in order to
avoid eventual catastrophic increases in
entropy right you have to seek it out
and master it
um and so that's why you need to be at
that
line between order and and chaos right
between order and
and disorder and you know as adam's
pointing out that's what
uh that's what evolved systems evolve
towards is toward the
edge of chaos because that maximizes the
possibility for computation it maximizes
the possibility for adaptation um
and there is
to make a segway from the abstract to
the more
creep because i feel like that's sort of
like what
my collaboration with adam it's like i'm
tying this to more concrete
psycho psychology he's trying to more
abstract theory
there's a lot of work in the last 20
years in personality psychology on
personality change and especially
personality trait change
one of the things we know from this
research is that virtually everyone has

at least one personality trait that they
would like to change
we also know that personality traits can
and do
change over time it might not always be
easy to change traits they might not
always change
you know dramatically but we do see
change in personality traits over time
we know that things like therapeutic
interventions change personality traits
especially reducing
neuroticism and personality
psychologists have recently been trying
to develop
uh systematic programmatic interventions
to help people to change
different personality traits and of
course a lot of it is through
first just changing more specific habits
more specific characteristic adaptations
because that's what as adam said you
have more
of an immediate grasp on is how to uh
how to change your habits how to change
your particular strategies
but it is possible for those to
generalize to these more
uh general changes and i would think of
that as you know you you learn specific
things but eventually those things that
you learn may actually tweak some of the
underlying general parameters that
actually enact those specific things
over time
thank you colin for those who need to
drop off
thanks for coming just to respect the

end of the official event but
blue dave and then we'll chill for a
couple more minutes but thanks everyone
so much for
coming today
um so i want to thank the authors
especially for for being here and
livening up this conversation but just
as far as things that i think we could
continue talking about or that would be
interesting to pursue
in the active inference framework um are
things like personality change or things
that do change the personality right
like trauma therapy psychedelics
mindfulness meditation like all of these
things can really have drastic effects
on who a person is
um fundamentally i i expect in like
terms of big five as well
and then also like maybe in a non-active
inference kind of way like the
discussion on consciousness
is is big and heavy and uh maybe not for
this live stream but
something that is definitely always
worth um talking about
thank you blue dave yeah
um sean carroll interviewed carl frist
in a few months ago
and he brought up i think the same point
that
professor de young just mentioned wait a
minute
we don't try to be certain about
everything we want novelty we want
we want to be confused and professor
christian became

even slower and more methodical and
self-critical and self-qualifying
than usual and really went i think was
working through all these levels of
mathematization
that i'm sorry in my humble opinion he's
kind of gotten trapped into
that's something that professor edelman
was terrified
as this incredibly brilliant young kid
and he's gonna get his brain
sucked by the algebra
thank you dave um you know it reminds me
it's controlled novelty
we can't just say more it's more as
different as what we say in complexity
and
when it comes to cybernetic goal-driven
edu chaos systems
it's not so simple as just take a linear
aggression and you know
more ants is better for the colony no i
mean selection has
clearly in different niches resulted in
colonies that have different
shapes of everything so it's a local
vector of selection but then the broader
space is actually not
quite so linear so i hear you on that
marco
yeah thanks i'm glad you mentioned that
podcast actually uh in my opinion
the best conversation with carl because
sean being also
professor carol also being a physicist
he really was able to
get carl um active and and it's more
playful and cheerful about talking about

it but i'm not sure
what you mean with the with the getting
more required and stuff critical i think
it's more
is awareness that that's really the the
main challenge in dealing with these
kind of seeming
tensions or or paradoxes because i do
think there's a very important
intention to point out when it comes to
formal modeling especially
when it comes to formally modeling
brains or humans
as we are all also familiar with um
which concerns
um as seen for example in the
personality model
and there are often these
complementarities these trade-offs
um and often in infor psychology in
normal
communication we tend to pick a trade or
pick a characteristic and just say
that's good or bad
it's always decontextualized it's rarely
in relation to the trade-offs
um in relation to which they obtain an
actual normative value
or valid normally value might be um
and crap something else oh yeah so um i
link to the youtube
on the youtube chat but but i really
personally am very fond of the theme of
pleasant surprises pleasant
perturbations
um and this controls
on these controlled unknowns for example
um well steven's gone now but but he

mentioned the idea of space
right so we could imagine um well first
should be more formal so formally um
that's
already resolved in 2017 um a paper i
was happy to be the co-author on
um because if active influence doesn't
concern itself with free energy per se
it's expected free energy right and the
more sophisticated your models
the more sophisticated your temporal
death and your contrafactual imagination
uh imagination is very formal but i but
i like
allowing us to talk about imagination as
if it was formal
right kind of liberating it's from
psychological connotations
and way too many centuries of philosophy
kind of dunking on it
right because it's simply because you
didn't have the grounding so
so it takes exactly an active influence
that these tensions
disappear if we allow ourselves to to to
accept
uh what the mathematics actually say but
there is this kind of trade-off but
there is this kind of complementarity
that indeed it is kind of the hidden
notion of yin and yang um and so it's no
surprise that stability and plasticity
is the meta trade in my opinion
um and oh yeah sorry and i linked uh
that paper from heinleisch
who is uh at the for university of
brussels and a pretty prominent research
ceremony

and he has a very interesting notion
which quite like about um
challenge propagation so um that's
continuing the theme of pleasant
surprises because
we design challenges for each other
social situations
games work all these cultural niches
our designs challenge is simply by
virtue of the fact that
they were selected by our social
cultural dynamics we allowed these to
persist
throughout time in culture knowing that
in their persistence they
enact positive effects on us as we try
to adapt to them
right and so i in that sense i find it
very very beautiful
um that these kind of controlled
unknowns is controlled chaos
is such a common theme in how we design
each other and allow each other to
cultivate and in that sense to another
theme
we co-regulate each other and in that
co-regulation of each other
we develop these kind of super personal
personalities which i think
well hopefully people can agree that we
sometimes really feel that viscerally
the kind of personality or vibe as as
millionaires
us like to say and there's some certain
vibes in aries
wow and these vibes tend to resonate
with our personalities
because we already have these models of

this kind of strategies and goals and
values we
we instantiate for particular situations
and so it's not far-fetched that the
same models
afford us also send certain
personality-like
themes in situations because that is
literally
the kind of brain system we use we use
the brain system of persons
to do this right the the brain system
somewhere
frontal there's a lot of performance
area
um we have next to each other a brainery
activated when we
do self-modeling and next to that
modeling other people
and so that is also exhibited in how
easily we attribute personality
or intention to inanimate super
simplified objects
right and that speaks to in my opinion
the validity of the core assumptions in
this
paper uh that personality can be
decomposed and indeed that's been
perfected of values strategies and goals
um sorry i went on a ramble i thought i
was going to say but but
yeah our relational architecture
the way that we talk about organic
chemistry and this chlorine wants to do
that and this galaxy
exactly that the intention the
intentional stance
that relational thinking is so deep and

the way that you put it with a super
personality it's like
the superman is the collective leviathan
it's not just the better
individual nest mate it's that and
in the context of something that's
actually super
or supra like above not just better
um scott and then anyone else who wants
to make a final comment but this has
been really
great marco this is i mean that was so
beautiful
he's starting to think of the
choreography of consciousness
you know that's what it felt like you
know it's such a beautiful notion
i always used to tell my kids now
they're in the 20s i used to say the
only time you see reality is when you
see paradox
and any other time you're inside a model
and the paradox and that goes right the
paradox is
the reality that's the reality of our
interaction you know you're interacting
then
you know you know you're in encountering
something
and so that notion of embracing the
paradoxical
and the dance and the and the
negotiation of it it reminds me the last
point i was going to make is it reminds
me of the research they did on fish
swimming they took a semi-rigid piece of
plastic and they put it in a
stream and they started increasing the

speed of the stream and at a certain
speed of the stream the plastic
undulated with a periodicity that was
akin to how fish swim and they realize
that fish don't swim through water they
pass through the water
and because they the water is causing
them
their swim similarly that choreography
of consciousness
is that right we're not we don't just
situate our
cognition we situate our persona and our
and and it's constructed externally and
internally
so the last thing i wondered about is is
each system gdelian
so each system is incomplete has
questions within it that can't be
answered from within it
and maybe our novelty seeking is an
intrinsic recognition that we're all
gdelians
that we need to go outside to find
answers
so anyway great discussion love it thank
you guys
thanks as always coming on
scott um so uh let's just go from uh
dave colin marco and then we'll probably
call it unless someone else has a
burning desire
the only time we need to be conscious is
where we're in that zone of paradox
we ought to understand it and we don't
nice colin well on the same theme
i really like that idea of paradox being
reality too and that's the same

as the prima materia of the alchemists
right it's what is the thing behind
everything
that gives birth to those things and
it's precisely the thing that we don't
understand

it's the thing that's unknown right and
so that's the chaos that's on the you
know the other side of
the border where we're at the edge of
chaos uh and yeah from
especially from the perspective of
consciousness what you need to be
conscious of is precisely only those
things that you
don't understand perfectly already
marco

yeah um uh yeah scott thank you that was
beautiful

um uh yeah i just wanted to riff on that
so so

i i laughed because um i mean i feel
comfortable saying this here so so
you know me and my friends we were
interested in psychedelics so
so i find it very very fascinating that
there's a huge variety
in what i call idiosyncratic encounters
so a lot of my friends with the
psychedelics have this one thing that
they feel like oh wow that was so
fundamental

and one of my friends uh let's call her
luta um

she had exactly that right she said i
couldn't escape the idea that everything
is parallax

right and uh though i won't go into it

but every other such quintessential
encounter
it is a great culture also in my opinion
relates to something
formalizable in terms of active
inference
um and i really really like the notion
of choreography of dance i i use the
metaphor a lot too
the um you might also be interested in
fridge of kaplan's
who also wrote the tao of physics and he
also had this moment
of insight about physics simply being
this uh
dance right he used a metaphor of god a
deity in hinduism
but more formally and interestingly well
my attempt to form
formally make it interesting is as we
noted right
cybernetics acts upon configurations of
models
simply because systems are composite or
compositional
and distributed and constantly
reorganizing so the very least can say
that it's about configurations
yes um going back to an earlier thread
if
conscious indeed is intermediating
cybernetics then they need to
intermediate the balancing or the
meta configuration of configuration
which themselves
are constantly changing so exactly in in
the example you took with the fish
they are embedded or immersed in fluid

dynamics

right there are flows and i say that

formally these are informational flows

so they're immersed in all these flows

and they need to find

or the space needs to find uh the space

need to situate itself

in between those flows and by virtue of

doing that

it is regulating them right and so i

think that's a very beautiful idea

um and the last thing i think is also

very beautiful so um we're all good

earlier

we're all in completes uh which reminds

me of conversations i've had with a

simon de deo a research at sfi institute

very very good researcher very

interesting has some very very nice work

on

bayesianism and the nature of knowledge

um

and and um i i think i said to him

exactly what you said right they saw

kadalian but but how i phrase it is

we're all each other's oracles so an

oracle and

you know in computer science but they

give you an answer that you can't

actually compute but if you would have

the answer then things would come easier

so it is by virtue of the noise we are

incomplete and that is allowed

because other people's incomplete

systems can act as an oracle for us

and so i think that's also something

that speaks to our experience of how

we aware certain things but we're it's

in in fact
actually very complete but we trust that
it's okay to be complete
and if interested in that then uh i'm
sure you know hofstadter goddess
terence deacon in complete nature as
adam mentioned
and the day as i mentioned so yeah
thanks so much that was very simulating
great thank you a bit interesting i
closed it with william blake quotation
he wrote in his notebook do what you
will this life's a fiction
and is made up of contradiction
they are beautiful i applaud to that
peace out everyone
thanks for participating thanks for
watching it hope to like
have more participation join the stream
if you're not on it already and
we'll see you next time thank you so
much for watching this
bye thanks everyone